

An Empirical Comparison of Parametric and Nonparametric Methods Applied to the Measurement of Election Fraud*

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Abstract

Two election forensics methods provide estimates of the magnitude of election fraud: Bayesian version of the finite mixture model developed by Walter Mebane and his students, and nonparametric approach based on vote-turnout histograms proposed by Sergey Shpilkin. While Mebane's approach offers rigorous statistical modeling based on the original Klimek model, Shpilkin's approach offers an intuitive and visually appealing algorithm. Using the data from Russian federal elections (2000-2021) this paper aims to compare both methods by conducting election forensics analysis and four validation studies from different elections. In addition, this paper comes up with two alternative methods based on univariate finite mixture models, as well as an algorithm designed to extract precinct-level estimates of election fraud from nonparametric method. The paper's main conclusion is that, despite the major differences in the approaches and identified biases, they can be relatively effective in measuring electoral fraud at both the precinct and aggregate levels.

Keywords: eforensics, nonparametric method, finite mixture model, Russian elections, precincts.

Introduction

Growing authoritarian tendencies around the world have increased general interest in studying elections marred by various irregularities and fraud. To date, modern science, namely the emerging field called *electoral forensics*, has developed few methods to effectively detect and measure election fraud.

New methods of election forensics provide valuable opportunities for scholars and practitioners interested in exploring electoral anomalies in various electoral settings. Among the most popular methods are the digit tests (Cantu and Saiegh 2011; Pericchi and Torres 2011; Mebane 2011; Beber and Scacco 2012; Kalinin and Mebane 2013) focused on the distribution of digits in electoral data, various 1D methods guided by normality or unimodality assumptions in detection of anomalies in turnout or vote shares (Myagkov, Ordeshook and Shaikin 2009; Shpilkin 2011; Kobak, Shpilkin and Pshenichnikov 2016), 2D methods based on correlation or bivariate regression analysis of turnout and vote shares (Sobyanin and Sukhovolsky 1995; Buzin and Lubarev 2008), and, finally, parametric methods relying on statistical modeling of election fraud (Klimek et al. 2012; Mebane 2016; Mebane et al. 2022).

Most of these methods fail to convey the key quantity of interest – magnitude of election fraud. Over the last ten years only two methods were able to the magnitude: the finite mixture model, specifically, its Bayesian version, called *eforensics*, developed by Mebane et al. (2022); and nonparametric method developed by Sergey Shpilkin (Kobak, Shpilkin and Pshenichnikov 2016). While *eforensics* offers rigorous statistical modeling based on Klimek’s original model, Shpilkin’s approach offers an intuitive and visually appealing algorithm popular among pundits and general public in the post-Soviet region. Observed discrepancies in the results of the finite mixture model and Shpilkin’s approaches, especially with respect to 2018 Russian presidential elections, drew criticism among some Russian observers (Kalinin 2018; Shpilkin 2018). Since then, *eforensics* replaced the finite mixture model by bridging the gap between the methods. However, as we shall see, the problem of a large discrepancy still persists.

Because election fraud is difficult to measure, validation studies that test the plausibility of conceptual ideas related to the measurement of electoral anomalies are essential. Unfortunately, no serious studies have yet been done to evaluate which method insures the best validity. Such a comparison has been difficult because of different scales (votes vs. probabilities) and levels (precinct-level data vs. aggregate-level data). The methods also differ in terms of underlying assumptions (unimodality assumption vs. normality assumption). At last, methodological differences between methods are overshadowed by the general problem of election forensics: the fundamental ambiguity of attributing the origin of anomalies to the results of normal politics (e.g., strategic behavior) or election fraud (Hicken and Mebane 2015; Mebane 2015).

In the past, most of the work focused on validation studies of each election forensics method separately. For example, Kalinin (2019) used the data from election observers, voting equipment and geographical data to validate the results from the finite mixture model for 2011 and 2012 elections. Mebane et al. (2022) used 32 real elections to validate electronic forensics measures derived from the *eforensics* model. To date, Kalinin and Mebane (2017) is the only study that attempted to provide a somewhat limited comparative analysis of the Shpilkin and Mebane approaches by assembling a small validation study based on the expert's region-level assessments of election fraud.

The purpose of this paper is to compare Shpilkin's and Mebane's methods by conducting election forensics analysis of Russian elections and performing four separate validation studies. This comparative analysis was made possible by replicating the Shpilkin method (the original algorithm and code are not in the public domain) and developing an algorithm to extract estimates of election fraud at the precinct level. In addition, this paper proposes two alternative methods based on the univariate finite mixture models, designed to provide alternative estimates. Unlike Shpilkin's model, based on straightforward algorithm and unimodality assumption for clean votes, these models are estimated using the expectation-maximization algorithm and rely on different distributional assumptions.

My election forensics analysis is based on the data from all Russian federal elections held between 2000 and 2021 (in particular, presidential elections in 2000, 2004, 2008, and 2018; parliamentary elections in 2003, 2007, 2011, 2016, and 2021; constitutional referendum in 2020). My validation studies use the 2018 data from the webcams, which is collected by the activists of the “For Fair Elections” movement; the parallel vote counting data for 2012, 2016, 2018, 2021 elections collected by SMS-CIK project; the voting equipment data for 2011 and 2012 elections provided by Sergey Shpilkin; and election observation data from Moscow 2011 parliamentary elections collected by “Citizen-Observer” (Enikolopov et al. 2013).

The structure of this paper is as follows. Section 1 provides an overview of election forensics methodology and description of three election forensics approaches for precinct-level estimation of election fraud: Shpilkin’s nonparametric method, univariate finite mixture models (precinct-based and histogram-based), and Mebane’s multivariate finite mixture models. Section 2 explores empirical strategy: hypotheses, data and methods. Section 3 discusses main findings of my empirical part by focusing on analysis of consistency of election forensics estimates across Russian federal elections and four validation studies for selected elections. In the final section I draw conclusions and discuss prospects for future research.

Measuring Election Fraud

Election fraud is an inherently opaque phenomenon. It involves clandestine efforts on the part of the perpetrators to shape election results in a desired direction (Lehoucq 2003). It can also be viewed as “a conduct intended to corrupt the process by which ballots are obtained, marked, or tabulated; the process by which election results are canvassed and certified” (Donsanto 2008). Although the hidden origins of fraud make it difficult to study, election forensics methodology suggests that election fraud may leave distinct traces in the electoral data that researchers intend to uncover. From a practical perspective, the difficulty of this task is compounded by data availability issues related to the lack of fine-grained data

and suitable covariates to build complex explanatory models of election fraud. Despite these challenges, election forensics methods can provide accurate information about election violations, designed to separate false accusations from true evidence of election malpractice, and thereby help election observers assess the quality of elections and understand the geography of election fraud (Hicken and Mebane 2015, 4).

One of the most popular methodological approaches are digit tests built on the comparison of empirical distribution of digits with a pre-specified theoretical distribution. For example, these are first digits of aggregate vote totals (Cantu and Saiegh 2011), second significant digits (Pericchi and Torres 2011; Mebane 2011), the last digits in vote counts (Beber and Scacco 2008) or last digits in percentages (Kalinin and Mebane 2013). While the first and second digits are expected to follow the Benford's Law (2BL) if elections are clean, last digits in vote counts or percentages are expected to follow a uniform distribution. By now the second digit test has been discounted as a valid test because of its susceptibility to false positives and the difficulty associated with disentangling strategic behavior from the effects of election fraud (Deckert, Myagkov and Ordeshook 2011; Mebane 2012; Mebane and Klaver 2015). Other approaches, such as, the last digits in vote counts (Beber and Scacco 2008) or last digits in percentages (Kalinin and Mebane 2013) are used by researchers as intuitively simple methods for fraud detection at different levels of aggregation.

An alternative methodological approach is to use density plots and correlation analysis to detect anomalies. For instance, using 1D methods we can create a 1-dimensional density plot displaying the distribution of turnout or vote choice. These methods are guided by normality or unimodality assumptions.¹ One the most popular methods within this family is a histogram-based approach (Shpilkin 2011; Kobak, Shpilkin and Pshenichnikov 2016), which will be discussed in greater detail in the next section. This method employs the 1D density

¹The distribution of turnout or vote choices should be approximately normally or unimodally distributed if the data are homogeneous and generated by uncorrelated factors. Under the normality or unimodality assumption, if some parts of electoral data are artificially changed, this would result in a non-normal or multimodal distribution, whereas in the absence of election rigging, the distribution should be approximately unimodally or normally distributed (Myagkov, Ordeshook and Shaikin 2009).

plot to determine the clean mode of the distribution and to extrapolate its information to the rest of the histogram. Other methods, like 2D methods, rely on correlation or bivariate regression analysis in determining the anomalous relationship between turnout and vote shares (Sobyanin and Sukhovolsky 1995; Buzin and Lubarev 2008). For instance, Sobyanin and Sukhovolsky (1995) argue that by regressing the winner's electoral support on turnout in a bivariate linear regression model, based on estimated beta coefficients, certain conclusions can be drawn about the fraudulent nature of elections.

Finally, parametric methods of election forensics are designed to model the underlying mechanism of election fraud (Klimek et al. 2012; Mebane 2016; Mebane et al. 2022). This approach relies on the Klimek et al. (2012) model, assuming that there is a winning party or candidate, i.e., the one with the most votes who benefits from the votes transferred from other parties/candidates and nonvoters. The original model consists of two components: the first component includes turnout and vote counts under condition of fair election (in this case, the vote counts are simulated from a normal distribution with the mean and standard deviation obtained from the area on the left-hand side of the mean); the second component accounts for the intensity of anomalies presumably associated with election fraud and is obtained from the right-hand side of the sample mean. In the works of Mebane (2016); Mebane et al. (2022), the Klimek model has been significantly improved with introduction of a finite mixture model using the expectation–maximization algorithm and, later, its Bayesian implementation, which is called *eforensics*.

Few forensic methods convey the key quantity of interest – magnitude of election fraud. Over the last ten years, two approaches have become popular: the aforementioned finite mixture model, and its modern version *eforensics*, and Shpilkin's nonparametric method (Mebane et al. 2022; Kobak, Shpilkin and Pshenichnikov 2016). While *eforensics* offers rigorous statistical modeling based on Klimek's original model, Shpilkin's approach offers an intuitive and visually appealing algorithm popular among the general public in the post-Soviet states. The observed discrepancies between both methods in the 2018 presidential

election led to criticism of the the finite mixture model by Russian bloggers and election observers, as the finite mixture model significantly underestimated election fraud: 320,000 vs. 10,357 million stolen votes obtained by the nonparametric method. After 2018 the finite mixture model was completely replaced by *eforensics*, which provided results closer to nonparametric approach (5,687 million).

Figure 1: Comparing Nonparametric (NP), Shpilkin's (original) and Mebane's (*eforensics*) methods for Russian federal elections and a referendum (2000-2021)



Notes: Bootstrapped standard errors. Statistically non-significant precinct-level anomalies are not excluded from the aggregates. Nonparametric method or *NP* is a replication of the original Shpilkin method.

It seems that the discrepancy problem observed in 2018 can be extrapolated to other Russian elections as well. Figure 1 shows two important observations: first, nonparametric method, that is, my replication of Shpilkin's method, is in close proximity to Shpilkin's assessments; second, the election-level aggregates for both methods differ significantly from the *eforensics* estimates.

Given the variability of election fraud assessments provided by different methods, it

is necessary to discuss the criteria for selecting the best one. According to Hicken and Mebane (2015, 4-5), ideally, a method for detecting electoral fraud should meet each of the following criteria: first, be able to avoid mislabeling problematic elections as problem-free, i.e. limit the number of false negatives; second, be able to avoid mislabeling clean elections as problematic, i.e. limit the number of false positives; third, be able to analyze electoral results in their entirety and at the most fine-grained level possible; fourth, be able to analyze geographic allocation of anomalies; fifth, be able to produce estimates of uncertainty that show our degree of confidence in conclusions. In sum, following Hicken and Mebane (2015), for a validation study, all election forensics methods should be capable of providing point and uncertainty estimates on the same scale, preferably, at the precinct level, and should be sufficiently accurate by limiting the number of false positives and false negatives.

In this section I focus on an overview of three main approaches towards estimation of election fraud: nonparametric approach, univariate finite mixture modeling approach and multivariate finite mixture modeling approach. All three approaches rely on a core multimodality assumption and the concept of clean mode. They also use the same data as inputs: the total number of eligible voters, the number of votes for the winner and the number of valid votes. Based on these inputs other observed quantities of interest can be derived, such as turnout, proportion of abstainers and winner proportion, etc.

Nonparametric Method

The growing need for clean and fair elections has increased public demand for election observation and analysis of electoral anomalies in Russia. The popularity of election forensics in the country was predetermined not only by public demand for clean elections, but also by supply of innovative methods designed to expose massive election fraud. Sergey Shpilkin, a physicist back in 2007 proposed his new nonparametric method based on vote-turnout histograms (Shpilkin 2011; Kobak, Shpilkin and Pshenichnikov 2016). By calculating the number of votes stolen by the winner, the method's creator provided estimates of election

fraud at the national or regional level for all Russian federal elections since 2000 (See Figure 1). Since full technical description and replication code are not available, following Kalinin and Mebane (2019), this paper presents its version of replication algorithm with an R-code posted on *GitHub*.

The replicated version of nonparametric method implies that the number of ballots received by each party is a function of turnout and the corresponding vote share at each polling station. The method uses a turnout-vote counts histogram (called *electoral matrix E*), where turnout (x-axis) is divided/binned into a number of intervals or bins (100 bins), within which the level of electoral support for each party or candidate is calculated (y-axis). Similar to the Klimek approach, this method relies heavily on the assumption of the highest left side mode, which is located in the region with moderate levels of turnout and considered to be clean. Unlike the Klimek model, however, this method does not rely heavily on the normality assumption, but rather on a unimodality assumption for clean elections. Moreover, the algorithm is not so much focused on the use of area under a mode curve as it is built on the idea of extrapolating information derived from the location of the highest mode to the rest of the histogram.

Finding the most appropriate “clean” mode is crucial to the success of this method. The algorithmic search for the clean mode in complex multimodal one-dimensional or univariate histograms is quite a challenge, so it is difficult to devise a search algorithm that effectively detects it. This problem is exacerbated in the case of disaggregated data (at the region or state level), where fewer observations contribute to excessive multimodality due to random noise, for which search algorithms simply cannot produce reliable results. Even at high levels of aggregation, the degree of separation between the modes must be accounted for: the distribution appears bimodal only when both modes have sufficient separation between them to suggest that the data were drawn from a mixture. In addition, the presence of bimodality does not always mean that the data were sampled from a multimodal distribution: strategic behavior or data heterogeneity unrelated to elections can affect this distribution.

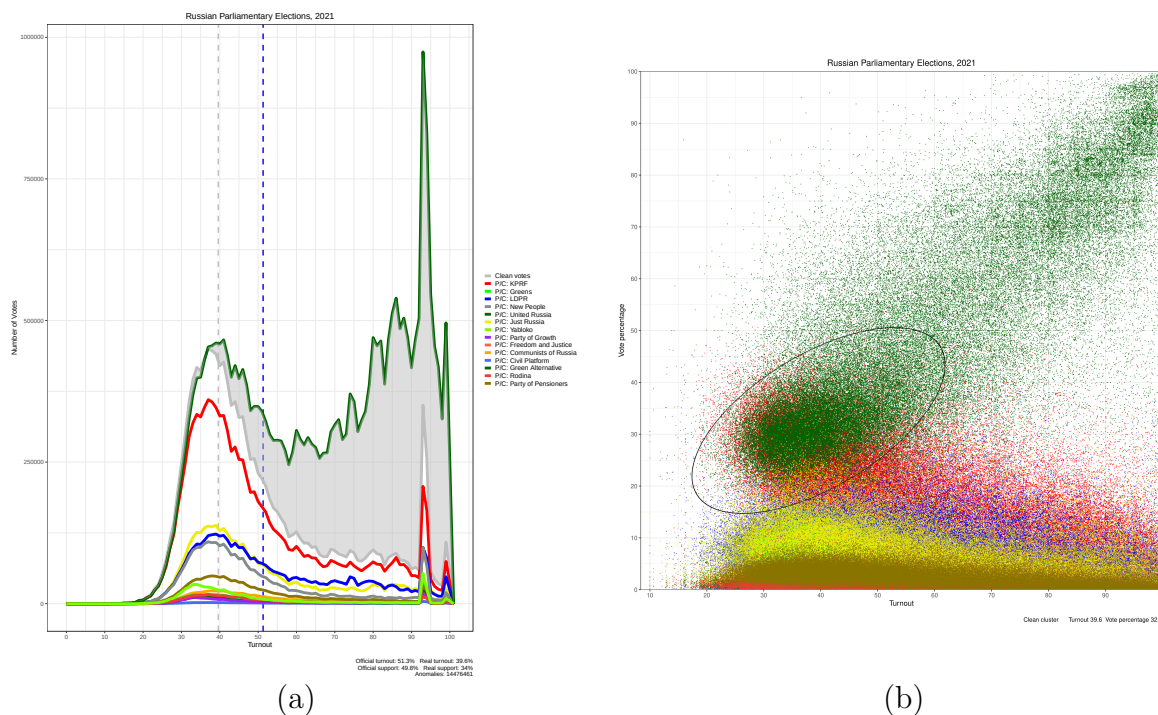
Some useful mode search algorithms proposed by researchers include, for example, the “peak probability contrast” technique, which provides a list of all common peaks among spectra, their statistical significance and relative importance in discriminating between the two groups (Tibshirani et al. 2004), the mvalue-based modal test based on the peak-search (Gregg 2021), lowland multimodality detection, based on studying the valleys rather than the peaks (Akinshin 2020). In R the *multimode package* written by Ameijeiras-Alonso, Crujeiras and Rodríguez-Casal (2021) is designed to detect the location of modes by estimation of the kernel density with Gaussian kernel. Another R package is *mixtools* written by Benaglia et al. (2009) uses standard expectation–maximization algorithm for mixtures of univariate normals that maximizes the conditional expected complete-data log-likelihood at each M-step of the algorithm.

In this paper I use two mode searching algorithms: a) an algorithm designed to find the highest mode on the left-hand side of the official turnout; b) an algorithm which finds the mode based on the largest “white” area of the rectangle, i.e. with the highest degree separation between modes. In case if all methods failed, manual selection of the clean mode can be considered as a viable option. In addition to the problem of random noise at lower levels of aggregation, another potential problem is that the degree of spikiness or smoothness of the distribution depends on the size of the bin, making clean mode detection somewhat troublesome. Also, this method can be fooled by cases with “artificial” unimodality of voter support for the winner at extreme levels of turnout. For example, in several cases this method shows that the estimated number of stolen votes in Chechnya, notorious for its election fraud, can be near zero or negative.

A more computationally efficient and reliable approach for finding local maxima would be to move from univariate to bivariate analysis. In particular, the problem associated with the clean mode search prompted Sergey Shpilkin to use a two-dimensional scatter plot, which visualizes the relationship between turnout and vote share by projecting precinct-level data for each party/candidate into two-dimensional space. It is suggested to use the minimum

covariance determinant (MCD) method, which is a highly robust estimator of multivariate location and scatter primarily used for outlier detection. MCD method allows us to estimate the covariance matrix, which can also help to calculate standard errors.

Figure 2: Visualization of Shpilkin's Approach



In MCD, α is a numerical parameter controlling the size of the subset over which the determinant is minimized. When $\alpha = 0.5$, about half of the $\alpha * n$ observations are used to compute the determinant (Maechler et al. 2021). In other words, the algorithm finds the covariance matrix with the lowest possible determinant, which is associated with the clean mode (Hubert, Debruyne and Rousseeuw 2018). The MCD method measures multivariate location and scatter of the clean data, while ignoring about 50% of the rest of the data regarding it as an outlier. The tradeoff between keeping α at 0.5 or increasing it is to possibly contaminate the “clean” cluster with areas that are anomalous in terms of turnout and vote shares, i.e. by increasing the number of false positive observations. Another approach is to use k-means clustering, designed to partition observations into a pre-specified number of

clusters. Determining the number of clusters can be difficult, especially given the problem of multimodality at lower levels of aggregation. Based on the Russian case, the choice of $k = 2$ or $k = 3$ is the most appropriate at the national level, but even small changes in the number of clusters can lead to variations in the “cluster centers,” which will impact the reliability of our final estimates.

Once the clean mode is identified, Shpilkin suggests using information about the ratio of votes received by the winner and all other candidates/parties to calculate the distribution of “clean” votes, assuming that the number of votes for all other parties is not a function of election fraud. Since the method is based on the premise that election fraud is a function of higher turnout and higher voting shares of the winner, it can effectively detect ballot stuffing rather than ballot switching.

To summarize, in its original version, the nonparametric method can be reproduced using the following algorithm (see Algorithm 1 for pseudo-code). First, an electoral matrix or histogram with a 1% bin for turnout (x-axis) is constructed, within which the number of votes for a given candidate/party is summed (y-axis). Second, the highest mode m to the left of the official turnout is determined, which is used to calculate the winner’s w clean vote share w_m , the real turnout level t_m (any mode search algorithm can be used at this stage), and, most importantly, the deflation factor d_m , which is the ratio between the winner’s clean votes and all valid clean votes minus the winner’s. Third, for each bin, the true electoral support for the winner is determined by weighting the votes for all other parties/candidates (except the winner’s party/candidacy) by d_m across all the bins of electoral matrix. Figure 2 shows anomalous votes as a shaded area, which is the difference between the area under the curve associated with the winner’s official electoral support and the area under deflated “theoretical” curve calculated using the weighting procedure. Following this logic, the magnitude of total election fraud can be calculated as $f_w = N_w - N_{all-w} \cdot d_m$, where: f_w is the number of fraudulent votes benefiting the winner, N_w is the number of votes received by the winner, N_{all-w} is the total number of votes received by all candidates/parties except the

winner.

Algorithm 1: Algorithm for Original Nonparametric Method

Input : Electoral matrix

$$E = \begin{bmatrix} v_{0,1} & v_{0,2} & \cdots & v_{0,c} \\ v_{1,1} & v_{1,2} & \cdots & v_{1,c} \\ \vdots & \vdots & \ddots & \vdots \\ v_{100,1} & v_{n,2} & \cdots & v_{100,c} \end{bmatrix}$$

where $i = 0, 1, 2, \dots, 100$, $j = 1, 2, \dots, c$. Each element v is a vote count for candidate/party j by turnout level i .

Input : t_o

Input : w

Output: f_w

1 $E_s = E_{i < t_o, j \geq 2}$

2 $V_w = E_{s, j=w}$

3 $m = \arg \max_{V_w \in \mathbb{R}} (V_w)$

4 $w_m = \frac{E_{s i=m, j=w}}{\sum_{j=1}^c E_{i=m, j}}; t_m = m; d_m = \frac{E_{s i=m, j=w}}{\sum_{j \neq w}^c E_{i=m, j}}$

5 $N_w = \sum_{i=0}^{100} E_{i, j=w}$

6 $N_{all} = \sum_{i=0}^{100} \sum_{j \neq w}^c E_{i, j}$

7 $f_w = N_w - N_{all-w} \cdot d_m$

One of the drawbacks of nonparametric method is that while it allows us to obtain aggregate measures of election fraud associated with the winner, they are not available at precinct level. The inability to do so prevents us from conducting any validation tests. This problem, however, can be solved by post-stratification that adjusts the precinct-level in such a way that the distribution of the winner's vote counts matches the known histogram-based distribution of clean votes (Kish 1965). Post-stratification is made by weighting the precinct-level vote counts, v_{it} , for each turnout stratum, t , by the ratio of the total number of clean votes in that stratum (histogram-based measure: $w_{ct} = \sum_{j \neq w}^c E_{i, j} \times d_m$) and the total number of votes for a winner in that stratum (precinct-level measure: $v_t = \sum_{i=1}^{N_t} v_{it}$). As a result of post-stratification, the sum of winner's votes in each stratum from the precinct-level data would be equal to the clean totals of the corresponding turnout stratum for the histogram data. By including an additional stratum, such as the *region* variable, groups need to be constructed as a complete cross-classification of *region* $\times$ *turnout* variables, so for simplicity

to allow multiple grouping variables we can use *raking* instead (Lumley 2010).

Procedurally, calculation of precinct-level quantities of interest for nonparametric method can be implemented using *svydesign* function of the *survey* package. Post-stratification is done with *postStratify* function to match precinct-level vote counts to the known distribution of clean votes in the electoral matrix. The obtained weights can be interpreted as precinct-level measures of clean counts; if $weights > vote\ counts$, I set $weights = vote\ counts$. In the end, the difference between the official winner's vote counts and clean counts provides a measure of anomalies at the precinct level. In our case with post-stratification weights based on only one stratifying variable, the weights can be easily calculated manually,² but using *svydesign* is more convenient for more than one stratification variables.

One of the problems associated with “disaggregation” of vote-turnout histograms to precinct level is that for some values of turnout, strata can have zero counts making any weighting impossible. One possible solution would be to either merge such strata with non-zero strata, or simply remove those empty weights from the analysis. I followed the latter solution by removing data with empty weights from my analysis.

Univariate Finite Mixture Models

A simple alternative to the nonparametric approach is to estimate a finite mixture model for a univariate electoral variable (for instance, turnout) using an expectation maximization algorithm. Univariate finite mixture models occupy an intermediate position between our nonparametric model and much more complex multivariate finite mixture models discussed in the next section (McLachlan and Peel 2000). Using a finite mixture model with a univariate density enables us, on the one hand, to replace the nonparametric algorithm with a simple model relying not so much on the multimodality assumption but rather on the mixture of normals, on the other hand, to use the entire area under the clean mode to make inferences

²Using this formula precinct-level fraud for precinct i belonging to turnout stratum t is calculated as $f_{it} = v_{it} - v_{it} \times \frac{w_{ct}}{\sum_{i=1}^{N_t} v_{it}}$.

about observation-level data. Even if this approach is ill-suited for complex multimodal distributions, it can be still effective especially for fitting leftmost “clean” mode satisfying the normality assumption.

A mixture model helps to represent the group structure in the data and model complex distributions by estimating components that accurately represent local areas that can be influenced by different data-generating processes. As a result, the mixture models can be used to describe different mechanisms behind observed multimodality.

Parametric formulation of mixture model can be as follows. Let the component density $f_i(y_j; \theta_i)$ belong to some parametric family, where θ_i , is the vector of unknown parameters for the i -th component density in the mixture. Then, the mixture density $f(y_j)$ with the g -component can be written as $f(y_j|\Psi) = \sum_{i=1}^g \pi_i f_i(y_j; \theta_i)$, where $\Psi = (\pi_1, \dots, \pi_{g-1}, \xi^T)^T$ is the vector with unknown parameters and $\pi = (\pi_1, \dots, \pi_g)^T$ is the vector of mixing proportions, such that $0 \leq \pi_i \leq 1$ and $\sum_{i=1}^g \pi_i = 1$ (McLachlan and Peel 2000).

Expectation-maximization (EM) algorithm is the most commonly used approach for fitting the mixture distributions and estimating the mixture model’s parameters. The EM algorithm seeks to find the MLE of the marginal likelihood by iteratively applying these two steps. Expectation step (E-step) requires the computation of the conditional expectation of $\log L_c(\Psi)$ given the observed data y using current iteration t of Ψ_t to compute Ψ , which takes the following formulation: $Q(\Psi, \Psi^t) = E_{\Psi^t} \{\log L_c(\Psi) | y\}$. The M-step on the $(t + 1)$ th iteration requires global maximization of $Q(\Psi, \Psi^t)$ with respect to Ψ to provide an update of Ψ^{t+1} , i.e. $\Psi^{t+1} = \arg \max_{\Psi} Q(\Psi, \Psi^t)$ (McLachlan and Peel 2000).

The component densities in the EM algorithm can have arbitrary component distributions, but in practice they are univariate or multivariate normal. In the univariate case the mixtures of normal distributions have solutions corresponding to local maxima that are consistent and asymptotically normal and efficient.

According to McLachlan and Peel (2000), computational problems when using mixture models arise from the unboundedness of the likelihood function in the presence of mixtures

of normal components with unequal variances in the univariate case or unequal covariance matrices in the multivariate case, for these cases the global maximizer does not exist. The imposition of homoscedasticity as a restriction can reduce the normal mixture-based solution (McLachlan and Peel 2000, 98). Another set of problems is related to selection of appropriate starting values for the EM algorithm and the choice of the number of components g in the mixture model. The presence of spurious clusters suggesting that observed multimodality is not necessarily due to multimodality *per se* but rather due to aggregation or ecological effects. The occurrence of spurious solutions is largely to the fact that the data are continuous, binned data as in the Shpilkin's approach helps to mitigate this problem (McLachlan and Peel 2000, 101). The possibility that the EM algorithm may converge to spurious cluster usually depends on specific starting point.

This study proposes two variants of univariate finite mixture models: a) univariate density built on precinct-level turnout ($UFMM_p$), and b) univariate density based on histogram's electoral matrix ($UFMM_h$) described in the previous section. Figure A4 in the Appendix (See rows: "Modes ($UFMM_p$)" and "Modes ($UFMM_h$)") illustrates density plots for a multi-component normal mixture obtained using maximum likelihood estimation to fit the model to the data. The observation-level mixing proportions based on the clean mode can be used to weight the winner's vote counts to derive the precinct-level number of clean votes.

Multivariate Finite Mixture Models

The first parametric model of election forensics, based on goodness-of-fit test was proposed by Klimek et al. (2012). The model consists of two components: a) distribution of turnout proportions under the condition of fair elections, which is simulated from a normal distribution with a mean and standard deviation obtained from the area on the left-hand side of the sample mean; b) distribution of anomalies obtained from the right-hand side of the sample mean. In Klimek's formulation, the model is conditional on the number of eligible voters and election fraud implies that votes are added to a winning party/candidate with some votes

being manufactured from nonvoters and some votes are stolen from the opposition. As a result, two types of fraud are estimated within a model: *incremental fraud* demonstrating moderate vote transfers (ballots switching); and *extreme fraud*, showing extreme cases of vote transfer (ballot stuffing). Two assumptions are behind the model: first, a winning party or candidate with most votes benefits from the votes transferred from other parties/candidates and nonvoters; second, turnout correlates with the magnitude of election fraud. The model has been tested on a subset of autocracies and democracies and showed its effectiveness at various levels of data aggregation.

More complex finite mixture modeling involves the inclusion of mixing proportions as functions of vector of covariates associated with the response. Following Klimek's model, Mebane (2016) suggests the finite mixture likelihood model with three distinct components: probabilities of incremental fraud, extreme fraud and no fraud for each precinct.

$$F(\mathbf{W}, \mathbf{O}, \mathbf{A} \mid \mathbf{N}; \Psi) = \sum_{j \in \{0, i, e\}} f_j \prod_{i=1}^n g_{jW}(W_i \mid N_i; \Psi) g_{jA}(A_i \mid N_i; \Psi) \quad (1)$$

In the model f_0 , f_i and f_e are probabilities of no fraud, incremental fraud and extreme fraud, where $f_0 + f_i + f_e = 1$. The model describes the joint density of the observed vote counts for the winning party W_i , the observed sum of votes cast for all other parties O_i and the number of observed nonvotes A_i , i.e. $(W_i; O_i; A_i)$ as being conditioned on the number of eligible voters in each precinct and estimated parameters $\Psi = (\alpha, \nu, \tau, \sigma_\nu, \sigma_\tau, \phi)'$ defined by Klimek's model. The estimated precinct-level parameters are α – intensity of incremental fraud, ν – the winner's vote proportion and its standard deviation σ_ν , τ – the turnout and its standard deviation σ_τ and θ – the proportion of anomalies. For maximum likelihood estimation Mebane (2016) uses an EM algorithm that helps to obtain conditional probability that each observation belongs to each of the three mixture components, i.e. $\hat{f}_{0i}$, $\hat{f}_{ii}$ and $\hat{f}_{ei}$. The likelihood ratio test is implemented to assess the improve of fit of the three-component mixture model *vis-a-vis* the one-component no fraud model. The key quantity of interest,

such as the number of votes produced by frauds is determined as $M_i = \sum_{i=1}^n N_i \hat{f}_{ii} + \xi_i$ and $M_e = \sum_{i=1}^n N_i \hat{f}_{ei} + \xi_e$, where ξ_i and ξ_e are expected proportions. While finite mixture model allows to obtain estimates of the conditional probability that each observation belongs to different mechanisms of election fraud, it doesn't provide the uncertainty of precinct-level probabilities.

Bayesian implementation of the finite mixture model (*eforensics* or *BFMM*) represents a generalization of the model proposed by Klimek et al. (2012) and Mebane (2016). The paper of Mebane et al. (2022) employs the familiar concepts of “incremental fraud” and “extreme fraud”. According to the model, the marginal probabilities of incremental, extreme and no fraud are π_1 , π_2 and π_3 , respectively. To address the problem of label shifting the prior sets π_1 to the largest value. As a result we get: $\tilde{\pi}_1 \sim U(0, 1)$; $\tilde{\pi}_2 \sim U(0, \tilde{\pi}_1)$; $\tilde{\pi}_3 \sim U(0, \tilde{\pi}_1)$.

Similar to Klimek et al. (2012), the model's inputs are the total number of eligible voters (N_i), the number of votes for the leader (W_i) and the number of valid votes (V_i). Based on these inputs, A_i , O_i , $t_i = V_i/N_i$ (turnout), $a_i = 1 - t_i$ (proportion of abstainers) and $w_i = W_i/N_i$ (leader proportion). The unobserved quantities include τ_i , the true turnout proportion; ν_i the true proportion of votes cast for the leader; ι_i^S and ι_i^M , the proportions of incrementally stolen from opposition or manufactured votes from abstainers, and v_i^S and v_i^M , the proportions of stolen or manufactured votes in extreme fashion. Also, Z_i is a random variable assigning case to specific mixture component, such that $Z_i = 1$ if no fraud occurred, $Z_i = 2$ if incremental fraud occurred, and $Z_i = 3$ if extreme fraud happened. One limitation of this setup is that the model does not allow the two types of fraud to be mixed.

As a result, the random vector of unobserved variables is $Y_i = (Z_i, \nu_i, \tau_i, \iota_i^S, \iota_i^M, v_i^S, v_i^M)$. Unobserved variables $\nu_i, \tau_i, \iota_i^S, \iota_i^M, v_i^S, v_i^M$ are conditioned on observed covariates and random effects via logistic function also capturing overdispersion or extra variation for unobserved parameters of interest at the observation-level. The aggregate-level quantities π_1, π_2 and π_3 for fraudulent turnout and fraudulent votes are obtained similarly to Klimek et al. (2012) by using different combinations of unobserved variable-proportions of ν_i, τ_i, ι_i^S (See Mebane

et al. (2022) for more details).

In *eforensics* the MCMC method is implemented using Metropolis-Hastings algorithm with four chains with the stopping rule of the estimated asymptotic variance of k_{th} component satisfying inequality $\sigma_{\theta k} < 0.05$.

The numbers of precinct-level fraudulent votes are calculated using the formulas: $F_{ti} = p_{ti}N_i$ for turnout and $F_{wi} = p_{wi}N_i$ for votes and for the aggregate totals: $F_t = \sum_{i=1}^n F_{ti}$ and $F_w = \sum_{i=1}^n F_{wi}$. Based on these quantities the total proportion of stolen votes in leader's favor $\psi_{wW} = F_w / \sum_{i=1}^n W_i$ or the proportion of manufactured votes $\psi_{tW} = F_t / \sum_{i=1}^n W_i$.

Hence, unlike the nonparametric method, the parametric model is built on the aggregation principle, first by estimating precinct-level fraud quantities and then by aggregating them to the election level. Moreover, in contrast to nonparametric methods and univariate finite mixture models, for which observation-level measures of uncertainty are derived with bootstrapping, the Bayesian framework provides credible intervals of observation-level quantities directly from the model.

In general, both parametric and nonparametric methods are based on the assumption of multimodality, which views the presence of bimodality or trimodality in the distribution of turnout and vote shares as statistical evidence of election fraud. This assumption may be false, since multimodality can also be generated by the normal features of politics, such as strategic voting. Since reliance on the multimodality assumption can potentially lead to false positives, the use of contextual and auxiliary information can be crucial in determining the origin of electoral anomalies in specific context.

Hence, all three approaches help us to extract observation-level quantities of interest: nonparametric approach derives precinct-level frauds using post-stratification adjustments to the winner's votes, univariate finite mixture model and *eforensics* based on a finite mixture model and expectation-maximization algorithm provide election forensics quantities by weighting the number of votes for a winner or the number of registered voters with their respective mixing proportions. While in the case of *eforensics*, precinct-level measures of

uncertainty are obtained directly from the model, for other forensics methods this can be done using parametric or nonparametric bootstraps.

Empirical Strategy: Hypotheses, Data and Methods

Although the described methods use different modeling and data estimation strategies, all three methods are based on the core assumption of *multimodality* originated from election fraud and the concept of a *clean mode*. Therefore, I expect that all these methods should be consistent with each other, if not completely, then at least partially. For each method: non-parametric (*NP*), univariate finite mixture model at precinct level (*UFMM_p*) and histogram level (*UFMM_h*) as well as *eforensics* (*BFMM*) I compute two precinct-level quantities of interest: a point estimate of election fraud (number of fraudulent ballots in the winner's votes) and a measure uncertainty (bootstrapped standard error). Following Rozenas (2017), I also estimate standard errors using resampled kernel density method for parametric bootstrap. According to the model for each precinct $i = 1, \dots, N$ with precinct-level turnout rates and support rates distributed according to unknown population-level distributions P_{t*} and P_{v*} , that is $t_i^* \sim P_{t*}$ and $v_i^* \sim P_{v*}$ we sample valid votes from the number of registered voters in precinct R_i with the success probability t^* : $T_i|t_i^* \sim \text{Binomial}(R_i, t_i^*)$; and the number of votes for the ruling party/candidate from $V_i|T_i, v_i^* \sim \text{Binomial}(T_i, v_i^*)$. Due to computer memory problems the number of bootstrap samples was limited to 100. Obtained bootstrapped precinct-level standard errors helped to determine statistical significance of precinct-level estimates. Statistically insignificant observation-level quantities of interest were assigned a null value and were excluded from my analysis (unless separately mentioned that statistically insignificant values were not excluded).

Consistency Hypothesis: Since all methods are based on a bi/trimodal turnout/vote distribution, one would expect relative consistency between derived estimates on any level of analysis. From a statistical perspective in this paper, *consistency* involves comparing between

different aggregates of election fraud, such as country-level estimates; and using correlation analysis or multivariate regression analysis with precinct-level quantities of anomalies to estimate the effects of different covariates. Consistency hypothesis can be tested for all Russian federal elections 2000-2021 at national and precinct levels.

Validation Hypothesis: Both parametric and nonparametric methods produce equally valid estimates of election fraud at any level of aggregation.

My *Validation Hypothesis* focuses not merely on the aggregates of election fraud, but also on the effects of key covariates (region's constitutional status, urban-rural divide, turnout, winner's vote share) on our forensics measures. The sign and magnitude of effects should be consistent with our theoretical expectations of how election fraud is organized in Russia.

In Russia election fraud is closely related to the strength of regional political machines. Hale (2003) argues that the ability of regional governors to convert patronage networks into political machines aimed to provide necessary level of incumbent's electoral support has been the case in the 2000s. The use of "administrative resources" implies that "different kinds of pressures are exerted upon voters by the regional authorities in order to provide desired electoral returns" (Goloso 2006). In Russia "political machines" are located in ethnic Republics and autonomous districts with dense ethnicity-based social networks (Hale 2007). Hence, here one would expect that precincts associated with Republics and regions with strong political machines are likely to be most fraudulent.

Election fraud is also expected to be associated with rural areas, where higher levels of social connectedness and lower levels of anonymity compared to urban area predispose to the use of local political machines. Reisinger and Moraski (2017) argue that local politicians with a prominent role in providing goods and services will be most successful in mobilizing voter support in their respected localities. For example, the literature suggests that rural social networks are effective in mobilizing voter support in Kyrgyzstan (Darr and Hesli 2010) and rural Africa (Kuenzi and Lambright 2011).

Finally, since a rational candidate uses vote maximization strategy, turnout and his vote

shares are expected to be always positively associated with election fraud.

Measuring election fraud is quite a challenging task, so separate validation studies are needed to test the plausibility of each election forensics method. In this paper I conduct several validation studies. The first validation study uses the 2018 data from the webcams collected by the activists of the “For Fair Elections” movement. The second validation study engages the parallel vote counting data for 2012, 2016, 2018, 2021 elections collected by SMS-CIK project. The third validation study draws on the data of the voting equipment for 2011 and 2012 elections. The fourth study focuses on election observation in Moscow 2011 parliamentary elections.

Finally, our election forensics methods can be divided into two groups: *aggregation methods*, in which key fraud quantities are estimated at the precinct level and then aggregated to a higher level to produce *aggregates*; and *disaggregation methods*, in which fraud quantities are estimated at the aggregated level and then disaggregated to the lowest level. The first group includes: *eforensics* and precinct-level univariate finite mixture model $UFMM_p$; the second group includes nonparametric method NP and histogram-based univariate finite mixture model $UFMM_h$. Thus, since the *eforensics* (or $BFMM$ is short for Bayesian Finite Mixture Model) approach involves aggregation of precinct-level fraudulent counts, and the nonparametric method, by contrast, relies on a disaggregation scheme, we would expect the latter to produce less valid results at precinct level. This is due to the fact that by resorting to extrapolation in disaggregation methods we are making improper inferences about units of analysis that were not actually analyzed. This can potentially lead to an aggregation bias in disaggregation methods compared to aggregation methods (Freedman 1999). This bias can be studied using “multi-level”, or “mixed” statistical models that take into account different, nested units of analysis. Following this logic we can formulate the following hypothesis:

Aggregation Hypothesis: The election forensics results from aggregation methods are expected to exhibit higher validity levels as opposed to disaggregation methods.

My general empirical strategy involves estimation of precinct-level anomalies using the

parametric and nonparametric methods discussed in the previous section. For each method, precinct-level point estimates and uncertainty estimates are obtained either at the estimation stage, as in the case of *eforensics*, or using parametric bootstraps, as in the case of other methods.³ To test my hypotheses: I compare election-level aggregates from different forensics methods, apply various statistical tools including correlation and multivariate regression analysis.

Findings

Consistency of Election Forensics Methods

Table 1 demonstrates an upward trend in election fraud from 2000 to 2021: *BFMM* estimates are half that of nonparametric methods *NP*, in turn *UFMM* estimates in certain years are twice that of *NP*. If we simply look at the intra-year correlations of various measures, we find a very strong correlation between *NP* and *BFMM* (0.88), a strong correlation between *NP* and *UFMM_h* (0.66), and the strongest correlation between aggregation methods *UFMM_p* and *BFMM* (0.94).

Correlation analysis of precinct-level data in Table 2 demonstrates several important findings. All precinct-level estimates are strongly correlated across the years. This is especially true for correlations between the estimates of *UFMM* and *NP*; *UFMM* estimates are also correlated with *BFMM*, but they are relatively weaker and inconsistent across the years. Finally, strong correlations are observed between *BFMM* and *NP* quantities are ranging between 0.70 and 0.9 with the only exception for 2021.

In addition to correlation analysis, Table 3 contains information about the overlap of our precinct-level measures. For example, *UFMM* methods and *BFMM* show from 62% in 2003 to 58% in 2021 of precincts are falling within bootstrap confidence intervals (*BFMM*'s

³Unfortunately, *BFMM* does not provide measures of uncertainty across all the elections, so these measures are excluded from my subsequent analysis.

Table 1: Aggregate Measures of Anomalies Across Elections

Year	Proportion of fraud in Winner's votes				Magnitude				Clusters
	NP	UFMM _p	UFMM _h	BFMM	NP	UFMM _p	UFMM _h	BFMM	
2000	0.063 (0.001)	0.192 (0.002)	0.165 (0.002)	0.033 (0.000)	2422089 (28847)	7376570 (68664)	6297518 (68168)	1277992 (18175)	2
2003	0.200 (0.000)	0.284 (0.001)	0.240 (0.001)	0.109 (0.001)	4556092 (10966)	6469612 (30450)	5617563 (28949)	2479084 (36332)	2
2004	0.147 (0.001)	0.217 (0.001)	0.230 (0.002)	0.085 (0.001)	7239609 (28575)	10701532 (71667)	23652521 (158728)	4202252 (41207)	3
2007	0.074 (0.407)	0.246 (0.002)	0.240 (0.002)	0.129 (0.001)	3302408 (18102508)	10953674 (71005)	24266829 (142767)	5753766 (47930)	3
2008	0.240 (0.031)	0.307 (0.002)	0.325 (0.002)	0.107 (0.001)	12554282 (1630710)	16034331 (104154)	33269128 (159833)	5607576 (41350)	3
2011	0.438 (0.001)	0.454 (0.001)	0.413 (0.001)	0.174 (0.002)	14173453 (21739)	14678377 (43315)	13567296 (49139)	5620956 (61201)	2
2012	0.228 (0.001)	0.305 (0.001)	0.291 (0.001)	0.104 (0.001)	10395887 (25008)	13913475 (62517)	13358836 (56677)	4751099 (51406)	2
2016	0.436 (0.000)	0.473 (0.001)	0.435 (0.001)	0.195 (0.002)	12435301 (11988)	13489848 (28668)	12548234 (28034)	5574427 (56210)	2
2018	0.165 (0.001)	0.307 (0.001)	0.294 (0.001)	0.101 (0.001)	9330469 (38669)	17321391 (57827)	16697754 (63693)	5686495 (49108)	2
2020	0.485 (0.007)	0.466 (0.002)	0.456 (0.002)	0.195 (0.001)	27645202 (395724)	26560626 (141253)	44389815 (46517)	11128566 (57872)	3
2021	0.517 (0.000)	0.573 (0.001)	0.519 (0.001)	0.191 (0.003)	14339192 (1368177)	15901551 (141253)	14438781 (46517)	5301265 (118521)	2

Notes: Bootstrapped standard errors in parentheses. Statistically non-significant precinct-level anomalies are excluded from the aggregates. Number of modes is estimated using *LaplacesDemon* package in **R**.

credible intervals are not available for all elections). Our comparison of the *NP* vs. *BFMM* shows an overlap in the 35 – 50% range (with the sole exception of the 2020 referendum with an overlap of 10%).

In addition to correlation and coverage analysis, I also compute the Wasserstein metric, which measures how much work is required to convert one distribution to another. To account for possible outliers, I use the Wasserstein distance of order $p = 2$. Table 4 shows the distances between each pair of precinct-level distributions. According to the table, *NP* estimates are consistently closest to *BFMM*: the distance is relatively small for fairly clean elections of 2000-2007, while it increases for the more fraudulent elections of 2008-2021.

Thus, the results produced by both methods fairly similar, but still differ in terms of

Table 2: Correlations between distributions (statistically significant)

	UFMM _p	UFMM _h	BFMM	UFMM _h	BFMM	BFMM
	vs. NP	vs. NP	vs. NP	vs. UFMM _p	vs. UFMM _p	vs. UFMM _h
2000	0.917	0.895	0.681	0.900	0.736	0.674
2003	0.952	0.960	0.829	0.946	0.864	0.817
2004	0.962	0.798	0.900	0.743	0.902	0.724
2007	0.923	0.899	0.883	0.737	0.916	0.716
2008	0.933	0.847	0.825	0.712	0.832	0.660
2011	0.959	0.957	0.757	0.971	0.798	0.778
2012	0.965	0.954	0.842	0.978	0.845	0.822
2016	0.965	0.970	0.801	0.980	0.806	0.794
2018	0.958	0.949	0.897	0.976	0.896	0.873
2020	0.931	0.967	0.827	0.826	0.826	0.757
2021	0.994	0.992	0.563	0.997	0.554	0.554

Table 3: Analysis of coverage

	NP vs. BFMM	UFMM _p vs. BFMM	UFMM _h vs. BFMM
2000	0.517	0.109	0.217
2003	0.616	0.147	0.285
2004	0.385	0.094	0.112
2007	0.468	0.145	0.137
2008	0.404	0.100	0.104
2011	0.325	0.066	0.266
2012	0.419	0.113	0.202
2016	0.491	0.083	0.242
2018	0.542	0.071	0.138
2020	0.092	0.153	0.050
2021	0.577	0.065	0.247

aggregate results and precinct-level quantities, with *BFMM* showing the most conservative estimates compared to the *NP* and *UFMM* approaches. The main explanation of this discrepancy will be given in the next section.

Regression analysis helps understand how key covariates affect election forensics measures. Table 5 contains the results of a linear regression with fixed effects and a linear mixed-effects model with inclusion of region and year random effects. The OLS part shows consistency in the signs of the effects across the models: *Turnout*, *Winner*, and *Republic* positively affect all our forensics measures, which is consistent with our theoretical expecta-

Table 4: Wasserstein distance between pairs of distributions (statistically significant)

	UFMM _p	UFMM _h	BFMM	UFMM _h	BFMM	BFMM
	vs.	vs.	vs.	vs.	vs.	vs.
	NP	NP	NP	UFMM _p	UFMM _p	UFMM _h
2000	95.18	80.33	29.54	37.01	118.37	98.61
2003	60.69	49.83	25.90	25.28	79.21	62.40
2004	92.67	245.02	79.67	179.18	169.02	310.52
2007	68.10	191.34	93.02	180.79	142.20	277.19
2008	119.77	305.33	112.70	230.58	225.69	413.13
2011	56.39	54.72	145.48	34.37	191.15	167.98
2012	94.14	92.36	116.33	26.02	203.49	193.77
2016	73.12	66.96	101.46	27.79	167.40	151.01
2018	174.76	167.76	75.07	26.59	243.94	233.76
2020	100.56	251.60	237.50	236.46	291.84	486.98
2021	200.39	197.62	407.50	31.98	592.15	585.56

tions (a positive effect of *Urban* areas is the only exception). The magnitude of effects varies across the models with *BFMM* showing smaller effects compared to other methods. The linear mixed-effects model is consistent with the OLS results, except for the nonparametric method, which unexpectedly demonstrates a negative effect of *Republics* on election fraud.

Thus, my *Consistency Hypothesis* focused on the relative consistency between methods is only partially confirmed. Although for most years, fraud totals between nonparametric and *eforensics* methods differ significantly, they consistently describe an increasing trend in election fraud over the years. Moreover, the signs of effects of our major covariates (*Republics*, *turnout*, *vote share*) for the most part are consistent with our theoretical expectations.

Validation Studies of Election Forensics Methods

Various auxiliary data sources on election observation, installation of electronic voting machines, webcams, provide valuable opportunities for validation of election forensics methods. The effects of observers, webcams and voting technologies on election frauds differ. For example, the presence of observers who subject election administrators to close scrutiny is likely not only to detect election irregularities, but also to prevent administrators from committing election fraud. Second, the installation of electronic voting machines at polling

Table 5: Consistency Study: Regression Analysis

	OLS				LME			
	NP	UFMM _p	UFMM _h	BFMM	NP	UFMM _p	UFMM _h	BFMM
Intercept	-565.809 (22.528)	-749.299 (28.377)	-751.175 (31.264)	-347.293 (14.336)	-14801 (83.45)	-16082 (109.392)	-12292 (123.37)	-5561 (50.077)
Turnout %	4.827 (0.224)	7.102 (0.311)	6.659 (0.324)	2.412 (0.108)	4.385 (0.018)	6.451 (0.024)	5.899 (0.027)	2.113 (0.011)
Winner's %	3.642 (0.205)	4.515 (0.244)	4.535 (0.288)	2.97 (0.146)	1.016 (0.016)	0.899 (0.022)	3.262 (0.024)	1.089 (0.01)
Republics	55.079 (9.745)	88.26 (13.261)	67.775 (13.947)	46.148 (6.51)	-704.341 (185.587)	2053.895 (243.248)	819.999 (274.215)	1439.887 (111.318)
Urban	80.909 (6.358)	101.223 (8.069)	125.274 (10.659)	36.039 (3.088)	61.36 (0.595)	79.871 (0.78)	114.196 (0.882)	21.474 (0.358)
Year 2003	119.866 (10.847)	138.437 (12.572)	144.807 (14.098)	83.334 (7.105)				
Year 2004	-9.663 (10.374)	-35.192 (12.954)	115.021 (15.598)	-18.154 (6.453)				
Year 2007	63.166 (10.905)	16.627 (12.653)	174.807 (16.378)	21.65 (6.947)				
Year 2008	21.264 (11.827)	-6.003 (15.729)	194.597 (22.478)	-11.182 (6.156)				
Year 2011	172.266 (12.221)	166.189 (13.631)	174.043 (14.454)	79.65 (7.895)				
Year 2012	55.01 (10.745)	38.878 (12.966)	50.366 (13.6)	9.112 (6.563)				
Year 2016	178.527 (12.112)	198.259 (14.858)	201.895 (15.315)	89.585 (9.142)				
Year 2018	-8.247 (12.101)	-1.84 (18.883)	10.162 (19.563)	-23.036 (8.166)				
Year 2020	166.458 (20.704)	92.214 (22.705)	299.528 (31.208)	31.623 (9.598)				
Year 2021	205.869 (15.026)	230.679 (19.435)	230.252 (19.76)	97.258 (8.641)				
R^2	0.245	0.244	0.237	0.245				
RE					3.59	3.71	2.85	1.26
u					246.99	323.99	366.17	148.55
LL					-7277913	-7562918	-7691431	-6743850
N obs	1050347	1050347	1050347	1050347	1050347	1050347	1050347	1050347

Notes: OLS – Ordinary Least Squares with fixed effects by year, clustered robust standard errors (year, region); Linear Mixed-Effects Models with random intercept (year, region).

stations can potentially change traditional means of manipulations, possibly leading to less fraud. Finally, installed webcams not only help expose election fraud by remote observers, but also influence the behavioral strategies of administrators, limiting the scope of traditional manipulation strategies.

In this part I explore whether precinct-level estimates of election fraud obtained by different methods can be validated by four datasets containing precinct-level data on election observation and voting technology. Specifically, the first dataset is based on voter counts from webcam recordings conducted during 2018 presidential election and was provided by Golos; the second dataset is obtained from the SMS-CIK website with the data on parallel vote counting for 2012, 2016, 2018 and 2021 elections; the third dataset contains the voting equipment: optical scan voting machines (KOIBs) and direct-recording electronic voting machine (KEGs) for 2011 and 2012 elections; the fourth dataset on election observation in Moscow 2011 elections coming from the Citizen Observer (Grajdanin-nabludatel) and obtained from the authors of Enikolopov et al. (2013).

Validation Study I: Counting Voters With Webcams

In this section I utilize the data from the March 2018 presidential election. The turnout data from the webcams for that election was collected by the activists of the “For Fair Elections” movement. Azat Gabdul’valeev from *Golos* organized data collection and data processing of this data. The independent observers were able to download footage from 8,000 polling stations out of 46,000 polling stations with installed webcams (Yurasova 2018; Golos 2018). The volunteers chose ten random polling places in each of Russia’s five regions with the highest turnout: Chechnya, Dagestan, Tatarstan, Karachay-Cherkessia, and Kabardino-Balkaria. The counting was conducted following the rule: every 15 minutes the number of voters casting ballots was reflected in a single-format report. While watching a video recording from selected precinct, volunteers recorded their answers to 11 questions describing the execution of procedures required by the law (Golos 2018). The quality check was done by

two unaffiliated volunteers who randomly checked 50 videos. In each of these cases, the results were nearly identical. A total of 271 reports/observations were used in this data analysis. The batches of reports came from different regions: Dagestan (31 reports), the Kabardino-Balkar Republic (16 reports), the Chechen Republic (12 reports) and the Kemerovo region (158 reports). Also, several reports were processed from Moscow city, St. Petersburg, Sevastopol, Kamchatskii krai, Karachay-Cherkessia Republic, Krasnodarskii krai, Pskovskaya oblast, Bashkortostan republic, Ingushetiya republic, Crimea republic, Mordoviya republic, North Ossetia-Alania republic, Tatarstan republic, Rostovskaya and Tyumenskaya oblasts.

Table 6: Validation Study I: Comparing Quantities of Interest

	Truth	NP	UFMM _p	UFMM _h	BFMM
Magnitude	97571 (5519)	125639 (5381)	238803 (10075)	233821 (10304)	89879 (4414)
Correlation	1 (0.000)	0.748 (0.037)	0.754 (0.026)	0.738 (0.030)	0.713 (0.035)
Wasserstein	0.000 (0.000)	139.896 (12.066)	619.360 (22.143)	615.403 (22.852)	129.556 (20.619)
Prop. fraud precincts	0.945 (0.014)	0.804 (0.024)	0.819 (0.023)	0.742 (0.026)	0.683 (0.028)

Notes: Bootstrapped standard errors in parentheses. Data for the 2018 presidential election.

Table 6 shows that compared to other methods, *BFMM* provides us with the most accurate estimate, whereas the nonparametric *NP* method provides the second best estimate. While the *BFMM* provides the estimate of 89879 with bootstrap confidence interval (81228; 94293) overlapping with the confidence interval of our true estimate of 97571 (86753.76; 108388.2); the *NP* measure of 125639 (115092; 136185) is biased upward and fails to overlap with the true confidence interval. There is one important caveat here: with *total fraud* *BFMM* outperforms other methods but also fails when *turnout-based fraud* is used (it is the only method that provides this quantity of interest). In the case of webcams, we are only interested in the latter indicator that amounts to 66253.49 (65864.91; 66642.07) and significantly lower the true estimate. To ensure comparability with other forensics methods, I've deliberately focused my analysis on total fraud, but for this validation study it is

important to mention both quantities of interest.

According to Table 6, among forensics methods, *BFMM* and *NP* are the closest to the truth in terms Wasserstein distances. Compared to *BFMM*, *NP* shows higher correlation and greater coverage of fraudulent precincts. It follows that while *BFMM* tends to produce more false negatives, *NP* produces more false positives. The density plots in Figure A2 in the Appendix visualize the coverage of our sample's true density by estimated densities: *BFMM* shows relatively better coverage of the true distribution than the nonparametric method.

This finding is supported by classification analysis provided in Table 7. It was implemented by binarization of all our precinct-level variables by coding “no election fraud” as “0” and the presence of election fraud as “1”. According to the Table, *NP* displays relatively higher accuracy (0.85) compared to *BFMM* (0.76). While in the case of the *NP sensitivity* is higher (0.87) compared to *BFMM*(0.75) (i.e. the method correctly identifies precincts with anomalies), the ability of nonparametric method *NP* vs. *BFMM* to detect precincts without fraud is comparatively low (0.69 vs. 0.81). This is not surprising, since *sensitivity* and *specificity* are inversely related.

Table 7: Statistics of Validation Studies

	Method	TP	TN	FP	FN	Acc.	Sensitiv.	Specif.	Prec.	Recall	F1
Study 1	NP	208	22	10	31	0.85	0.87	0.69	0.95	0.87	0.91
	UFMM _p	206	24	8	33	0.85	0.86	0.75	0.96	0.86	0.91
	UFMM _h	193	24	8	46	0.80	0.81	0.75	0.96	0.81	0.88
	BFMM	179	26	6	60	0.76	0.75	0.81	0.97	0.75	0.84
Study 2	NP	171	7313	1711	736	0.75	0.19	0.81	0.09	0.19	0.12
	UFMM _p	104	7868	1156	803	0.80	0.11	0.87	0.08	0.11	0.10
	UFMM _h	91	8028	996	816	0.82	0.10	0.89	0.08	0.10	0.09
	BFMM	23	8741	283	884	0.88	0.03	0.97	0.08	0.03	0.04
Study 3	NP	7350	76748	103014	2703	0.44	0.73	0.43	0.07	0.73	0.12
	UFMM _p	7083	71698	108064	2970	0.42	0.70	0.40	0.06	0.70	0.11
	UFMM _h	7568	61492	118270	2485	0.36	0.75	0.34	0.06	0.75	0.11
	BFMM	9236	37537	142225	817	0.25	0.92	0.21	0.06	0.92	0.11
Study 4	NP	17	1666	1454	26	0.53	0.40	0.53	0.01	0.40	0.02
	UFMM _p	19	1425	1695	24	0.46	0.44	0.46	0.01	0.44	0.02
	UFMM _h	9	2248	872	34	0.71	0.21	0.72	0.01	0.21	0.02
	BFMM	0	3013	107	43	0.95	0.00	0.97	0.00	0.00	—

The panels with bivariate linear regression lines depicted in Figure A1 in the Appendix show election forensics methods demonstrate strong predictive power: the slope coefficient is statistically significant and positive ranging from 0.39 to 0.75 and R -squared ranges from 0.51 to 0.56. Most of precincts are clustered around the regression line.

My regression analysis in Table 8 shows that the effects of *Turnout*, *Winner*, *Urban* and *Republics* on the ground truth are largely consistent with the effects associated with election forensics models, thus suggesting that forensic election methods can indeed capture the variance-covariance structure of data pretty well.

Finally, our *Aggregation Hypothesis*, suggesting that aggregation methods are expected to have higher validity compared to disaggregation methods, can be rejected. Even though an aggregation method, such as *BFMM*, shows the most valid results as opposed to other methods, we do not have enough evidence to conclude that one group is better than the other.

Validation Study II: Parallel Vote Counting with SMS-CIK project

The second dataset used for validation is obtained from the website of the SMS-CIK project. The idea of SMS-CIK is to allow observers to conduct parallel vote counting by promptly reporting information from the precinct-level protocols. The procedure is rather simple: the phone messages containing the figures from precinct-level protocols are texted by observers to a specific phone number, then they are processed by the servers and instantly become publicly available through the website *www.sms-cik.org* and are updated in real time. The published data on voting results are based on SMS messages received from observers accredited by the SMS-CIK project. Data from non-accredited numbers are accepted, but are less reliable and are used for comparison purposes. Unfortunately, this procedure implies that election observers are self-selected into the database rather than randomly assigned to precincts by design. Since 2012 the project has collected data on more than 1200 local, regional, and federal elections.

Table 8: Validation Study I: Regression Analysis

	OLS					LME				
	Truth	NP	UFMM _p	UFMM _h	BFMM	Truth	NP	UFMM _p	UFMM _h	BFMM
Intercept	-980.01 (308.616)	-1659.84 (115.674)	-3103.42 (209.809)	-3130.48 (206.106)	-1642.78 (149.457)	-867.03 (179.778)	-1659.84 (122.381)	-3103.42 (230.207)	-3130.48 (262.114)	-3130.48 (262.114)
Turnout %	13.632 (2.035)	17.646 (1.863)	30.697 (3.402)	29.102 (4.433)	8.445 (0.901)	13.952 (1.98)	17.646 (1.397)	30.697 (2.628)	29.102 (2.992)	29.102 (2.992)
Winner's %	0.929 (5.218)	7.099 (3.228)	15.851 (5.899)	17.447 (6.663)	14.73 (2.71)	-0.412 (2.997)	7.099 (2.109)	15.851 (3.968)	17.447 (4.518)	17.447 (4.518)
Urban	100.906 (37.469)	88.753 (27.539)	164.197 (68.919)	167.403 (87.697)	43.524 (19.802)	88.041 (44.514)	88.753 (31.92)	164.197 (60.044)	167.403 (68.366)	167.403 (68.366)
Republics	231.845 (56.343)	64.398 (15.588)	64.823 (30.71)	62.108 (37.356)	13.299 (22.831)	204.251 (89.334)	64.398 (24.334)	64.823 (45.775)	62.108 (52.119)	62.108 (52.119)
R^2	0.433	0.707	0.692	0.626	0.694					
RE						108.07	0	0	0	30.07
u						235.05	182.01	342.37	389.83	144.09
LL						-1862	-1780	-1948	-1983	-1725
$Nobs$	271	271	271	271	271	271	271	271	271	271

Notes: OLS – Ordinary Least Squares with fixed effects by region and year, clustered robust standard errors (year, region); Linear Mixed-Effects Models with random intercept (year, region).

For my study, the data is available for 2012, 2016, 2018, 2021 federal elections. The key variable of interest is “control ratio”, which indicates whether discrepancy is detected between observer inputs and official electoral results (“yes”, if no discrepancy and “no”, if there is discrepancy). SMS-CIK project warns that the observed mismatch could be caused by either simple typos of observers or election fraud. Although the “control ratio” is a somewhat noisy indicator of election fraud, it can still serve as a proxy indicator.

For this validation study, Table 7 demonstrates higher accuracy of *BFMM* compared to *NP* (0.88 vs. 0.75), higher specificity (0.97 vs. 0.81), but lower sensitivity (0.03 vs. 0.19).

Table 9: Validation Study II: Regression Analysis

	OLS				LME			
	NP	UFMM _p	UFMM _h	BFMM	NP	UFMM _p	UFMM _h	BFMM
Intercept					-82.343 (9.681)	-161.636 (16.052)	-129.142 (15.65)	-55.288 (5.672)
Turnout %	5.358 (0.614)	8.217 (1.013)	6.116 (0.964)	2.415 (0.343)	3.735 (0.093)	5.818 (0.144)	4.318 (0.145)	1.66 (0.059)
Winner's %	2.519 (0.389)	2.47 (0.515)	2.277 (0.5)	1.106 (0.25)	-1.151 (0.076)	-1.612 (0.118)	-0.89 (0.118)	-0.35 (0.048)
Control ratio	7.38 (3.556)	12.037 (5.556)	10.364 (5.206)	3.105 (2.335)	12.721 (3.293)	21.347 (5.11)	17.391 (5.142)	5.972 (2.106)
Urban	32.977 (7.521)	38.102 (11.667)	27.003 (11.594)	11.465 (4.291)	-0.011 (3.122)	-1.84 (4.848)	-3.386 (4.877)	-1.874 (1.996)
Year 2016	141.256 (14.76)	192.573 (23.628)	145.03 (23.614)	61.728 (9.24)				
Year 2018	-64.241 (8.704)	-52.484 (12.832)	-42.25 (12.803)	-21.597 (6.361)				
Year 2021	153.821 (17.743)	207.197 (26.956)	158.482 (26.354)	69.186 (10.833)				
R^2	0.459	0.431	0.344	0.291				
RE					43.95	0.04	0.04	0.01
u					93.1	144.48	145.37	59.57
LL					-59251	-63620	-63677	-54809
$Nobs$	9931	9931	9931	9931	9931	9931	9931	9931

Notes: OLS – Ordinary Least Squares with fixed effects by region and year, clustered robust standard errors (year, region); Linear Mixed-Effects Models with random intercept (year, region).

Table 9 illustrates that the effect of CR (“control ratio”) on election fraud is positive

and statistically significant for all the methods (*BFMM* fails to demonstrate statistically significant results in the OLS model). Hence, this study, with some reservations, validates all of our election forensics methods.

Validation Study III: Electronic Voting Machines and Election Fraud

The adoption of new voting or election observation technologies tends to affect the behavioral strategies of election administrators. The impact of technology on electoral behavior has been studied by Herron (2010), Sjoberg (2014), Bader (2013). Installation of electronic voting machines in selected precincts makes traditional manipulation, such as ballot stuffing or ballot switching, more costly for election administrators, subsequently resulting in a reduction in electoral malfeasance. Both approaches work well under the assumption that precincts are randomly assigned to receive treatment or not, so that randomization balances out possible confounders across treatment and control groups. The voting equipment data on both elections has been kindly provided by Sergey Shpilkin.

In both 2011 and 2012 elections two types of new voting machines has been used. The first, KOIB (Kompleks Obrabotki Izbiratelnykh Byulletenei), is an optical scan voting machine with a ballot box made of translucent materials. As one of its features, it creates a time-stamped file of the scanned ballot, and transmits election results over telecommunications network to a higher level of election commission (it's an integral part of the GAS Vybory automated tabulation system). The second, the KEG (Kompleks Elektronnogo Golosovaniya) is a direct-recording electronic (DRE) voting machine with a touch-screen with lists of candidates and parties for the voter to choose from. Its results are stored in the memory of KEG, as well as printed on a special paper tape unavailable to the voter. Both devices can induce election administrators to abstain from traditional election fraud, such as ballot stuffing and ballot switching.

Hence, I expect that more vote rigging is taking place in the polling stations equipped with the ballot boxes rather than KOIBs/KEGs.

Table 7 demonstrates higher accuracy of nonparametric measures compared to *BFMM* (0.44 vs. 0.25), higher specificity (0.43 vs. 0.21) and lower sensitivity (0.73 vs. 0.92).

Table 10: Validation Study III: Regression Analysis

	OLS				LME			
	NP	UFMM _p	UFMM _h	BFMM	NP	UFMM _p	UFMM _h	BFMM
Intercept					-195.175 (19.805)	-326.681 (24.014)	-263.314 (23.746)	-146.258 (12.966)
Turnout %	4.005 (0.325)	6.128 (0.456)	4.779 (0.435)	2.024 (0.171)	4.437 (0.041)	6.555 (0.05)	5.139 (0.051)	2.267 (0.026)
Winner's %	2.585 (0.315)	2.742 (0.324)	2.669 (0.293)	2.057 (0.173)	0.712 (0.035)	0.894 (0.043)	1.116 (0.044)	1.003 (0.023)
Urban	73.51 (12.442)	93.949 (16.166)	80.89 (16.484)	30.774 (5.704)	38.802 (1.23)	59.707 (1.49)	52.085 (1.533)	11.237 (0.789)
KOIB	5.932 (5.341)	7.305 (6.628)	5.55 (6.616)	7.478 (2.192)	-1.311 (2.086)	0.167 (2.527)	-0.446 (2.6)	3.403 (1.338)
KEG	-122.594 (72.836)	-148.38 (75.664)	-152.89 (83.649)	-46.417 (34.349)	-121.283 (8.213)	-147.066 (9.949)	-151.775 (10.238)	-45.658 (5.267)
Year 2012	-96.55 (8.932)	-95.256 (8.777)	-80.138 (7.608)	-54.326 (4.506)				
R^2	0.444	0.469	0.409	0.447				
RE					0.08	0.08	0.08	82.4
u					197.92	239.78	246.73	126.93
LL					-1273390	-1309804	-1315224	-1189075
$Nobs$	189815	189815	189815	189815	189815	189815	189815	189815

Notes: OLS – Ordinary Least Squares with fixed effects by region and year and clustered robust standard errors (year, region); Linear Mixed-Effects Models with random intercept (year, region).

Table 10 shows that the effects of KOIBs on election fraud are statistically insignificant for all our methods and model specifications, except *BFMM*, for which it is positive, which contradicts our intuition about the effects of KOIBs on election fraud. The effects of KEGs on election fraud are negative, which supports our expectation that less vote rigging can be taking place in the polling stations equipped with KEGs.

In sum, our findings for this validation study are very mixed: on the one hand, none of the forensic methods seem to meet our theoretical expectations for KOIBs, but all seem to satisfy our expectations regarding KEGs.

Validation Study IV: Election Observation in Moscow, 2011

For the purpose of this validation study, I use a data set on election observation in Moscow previously analyzed in Enikolopov et al. (2013). The data originally comes from the independent nongovernmental organization Citizen Observer (Grajdanin-nabludatel), which trained more than 500 volunteer observers in the city of Moscow. The observers were sent to 156 randomly selected polling stations using a systematic sampling technique. Enikolopov et al. (2013) conclude that the mere presence of observers reduced the vote share for United Russia by 9.3-10.8 percentage points and increased it for other parties.

For this validation study, Table 7 shows that higher accuracy of *BFMM* compared to *NP* (0.95 vs. 0.53), higher specificity (0.97 vs. 0.53) and lower sensitivity (0.00 vs. 0.40). Thus, as we have seen before, *BFMM* is much better at predicting precincts with no fraud than *NP*.

According to Table 11, the negative effect of “Observers present, serious violations reported” on our fraud estimates may make sense, especially if the commissions acted on the violations found. However, the positive effect of “Observers present, no serious violations” contradicts our theory, since we expect the presence of observers to yield negative effect on election fraud. Finally, the negative effect of “observers present, no protocol received” on election fraud also contradicts our theory, since the absence of a protocol is quite likely related to election fraud.

Hence, although *eforensics* and nonparametric method seem to be remarkably effective in detecting election fraud, especially, in our first two validation studies, the last two validation studies provide us with more ambiguous results: neither method shows superiority over the others.

Table 11: Validation Study IV: Regression Analysis

	OLS				LME			
	NP	UFMM _p	UFMM _h	BFMM	NP	UFMM _p	UFMM _h	BFMM
Intercept					-1016.113 (17.13)	-1210.115 (22.06)	-1015.782 (25.149)	-213.16 (12.742)
Turnout %	16.072 (0.906)	20.352 (1.003)	17.024 (1.115)	2.995 (0.504)	16.696 (0.336)	20.685 (0.432)	17.267 (0.493)	2.808 (0.256)
Winner's %	5.684 (0.536)	3.807 (0.713)	2.807 (0.749)	1.694 (0.43)	5.222 (0.236)	3.361 (0.304)	2.329 (0.347)	1.459 (0.177)
Observers present, serious violations reported	-33.077 (21.644)	-34.776 (27.076)	-24.371 (29.435)	-1.709 (8.729)	-31.605 (22.718)	-34.359 (29.105)	-24.865 (33.362)	-7.821 (18.013)
Observers present, no serious violations	58.006 (12.269)	85.89 (17.163)	86.966 (17.654)	34.059 (6.533)	59.149 (17.549)	86.535 (22.48)	87.118 (25.771)	31.971 (13.933)
Observers present, no protocol received	-32.876 (27.614)	-32.794 (26.02)	-32.981 (28.626)	-13.715 (8.265)	-39.524 (24.126)	-42.397 (30.909)	-42.789 (35.43)	-14.893 (19.13)
R^2	0.761	0.69	0.536	0.195				
RE					39.05	52.68	57.19	14.2
u					143.89	184.22	211.31	115.23
LL					-20311	-21096	-21524	-19555
$Nobs$	3163	3163	3163	3163	3163	3163	3163	3163

Notes: OLS – Ordinary Least Squares with fixed effects by region and year and clustered robust standard errors (year, region); Linear Mixed-Effects Models with random intercept (year, region).

Conclusion

Overall, this study finds that both of our methods, i.e. *eforensics* and nonparametric, despite significant differences between them, can be very useful in election forensic analysis. In particular, the study shows that *eforensics* method, by minimizing false positives and maximizing false negatives, produces more conservative results compared to nonparametric method, which maximizes false positives and minimizes false negatives. According to one of my validation studies, the results from nonparametric method can be inflated by at least 30%, whereas the results of *eforensics* depending on the variable we choose, can be either unbiased or, in the case of turnout-based fraud, can be deflated by about 40%. Presumably, direction of each method's biases need to be accounted for during election forensics analysis. The results also suggest that *eforensics* by using an aggregation approach may be better suited for analyzing observation-level data than a nonparametric method based on a disaggregation approach. However, there is no consistent evidence to suggest that aggregation methods perform strictly better than disaggregation methods.

In the future, I hope to focus on the analysis of subnational units (regions, territories) and conduct geographic cluster analysis using various election forensics quantities. I also hope to expand this study by bringing in cross-national election data and adding more data for validation studies to better understand the pros and cons of each election forensics method.

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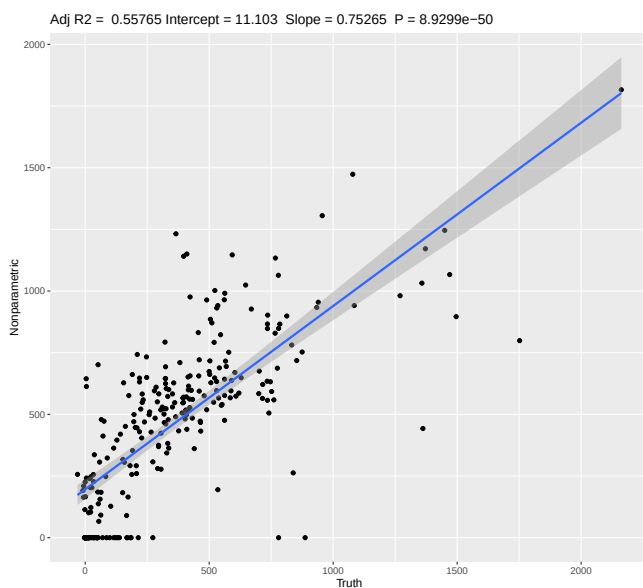
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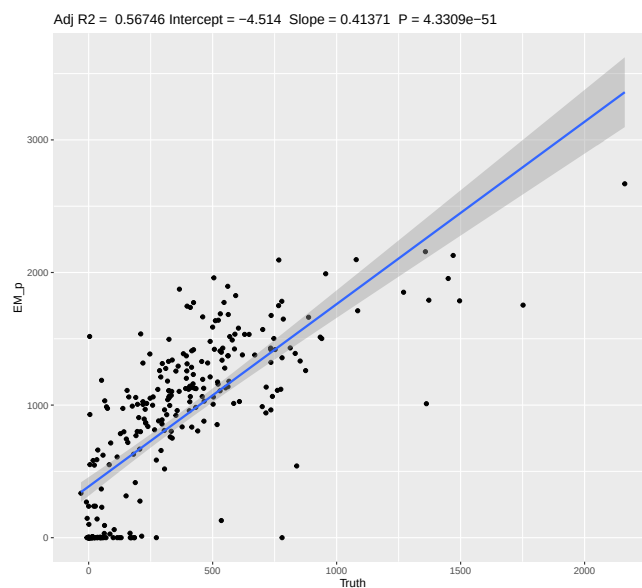
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A Appendix. Supplementary Tables and Figures

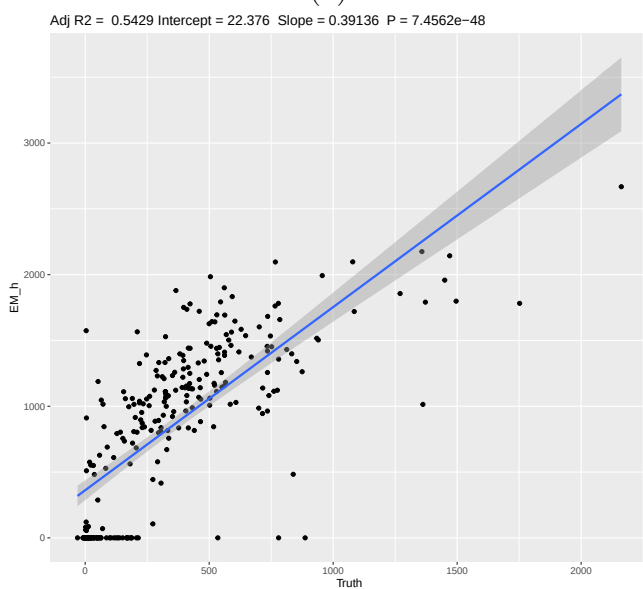
Figure A1: Least Squares Regression in Validation Study I



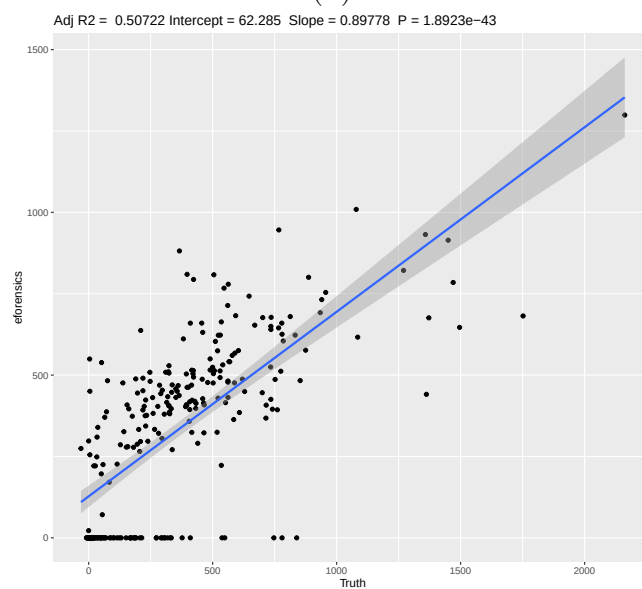
(a)



(b)

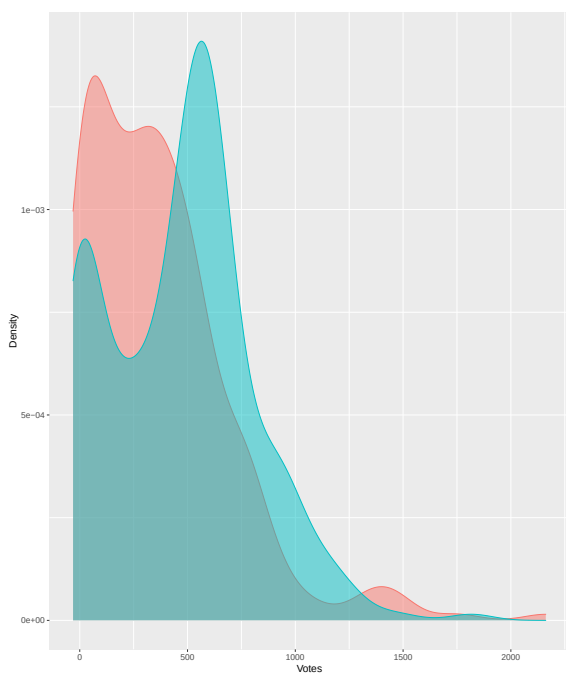


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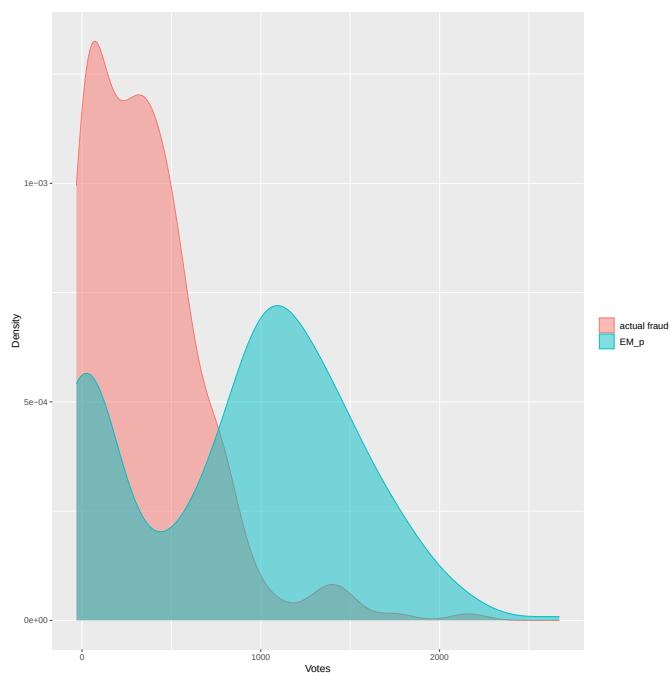


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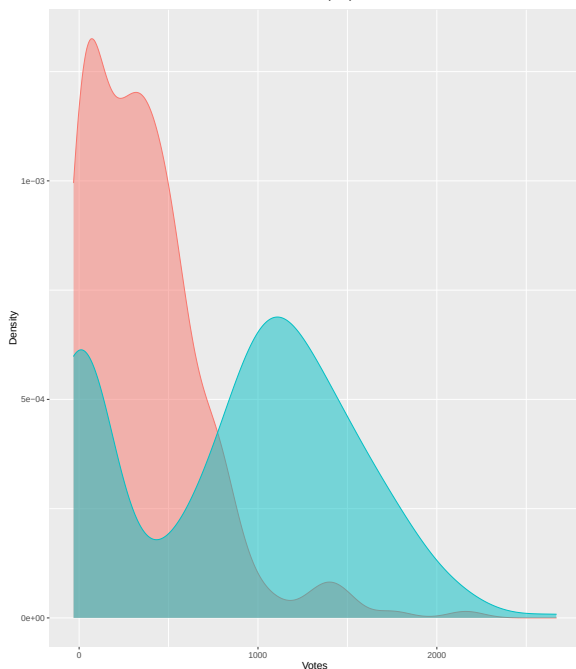
Figure A2: Density Plots in Validation Study I



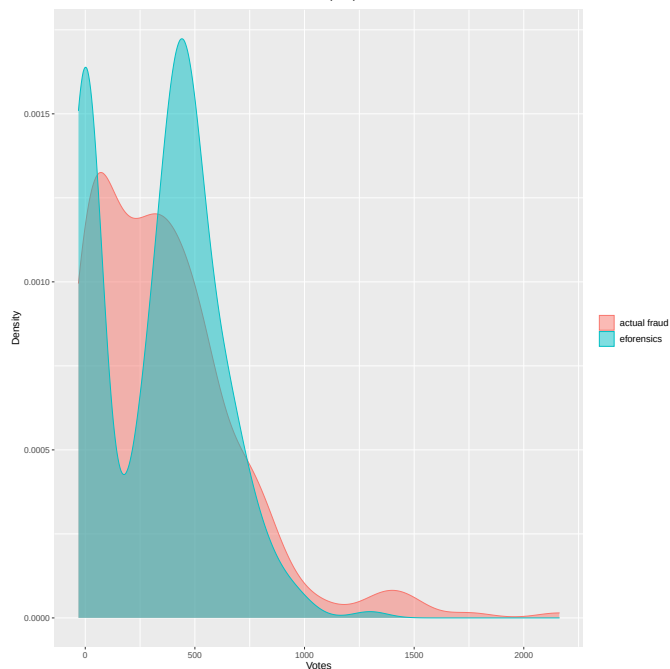
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(b)



(c)



(d)

Figure A3: 1D and 2D Density Plots for Different Election Forensics Methods

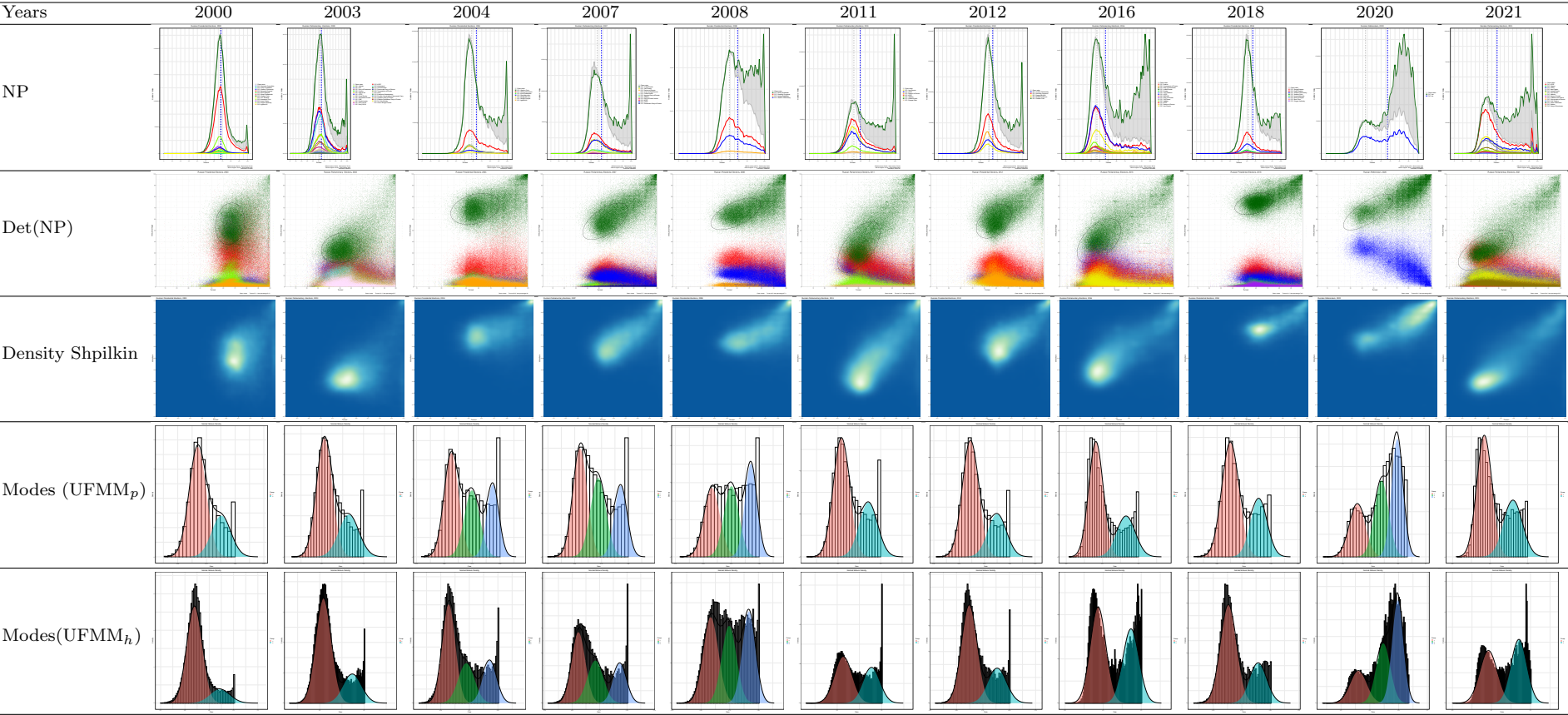


Figure A4: Fraud Densities by Method and Year

